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Paper Id: 238135 **Roll No.** | | | | | | | | | | |

B.TECH.
(SEM IV) THEORY EXAMINATION 2022-23
ENERGY SCIENCE AND ENGINEERING

Time: 3 Hours ***Total Marks: 100***

Note: 1 Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

(a) Give the examples of Potential Energy.
(b) What is energy quantization?
(c) If you turn on 4 light bulbs, each rated at 40 w, how long can they be on before you reach 1kwh?
(d) What are the two types of charge carriers in semiconductors?
(e) How is energy generated from bio energy?
(f) What is the anticipated life of a wind farm?
(g) Why is ocean thermal energy conversion a renewable resource?
(h) Write the characteristics of a green building.
(i) What is the criteria for LEED rating ?
(j) How do you calculate the embodied energy of material?

SECTION B

2. Attempt any *three* of the following: 10x3=30

(a) Derive time independent Schrodinger wave equation.
(b) Calculate the binding energy per nucleon of $_{17}Cl^{35}$ nucleus. Given Mass of $_{17}Cl^{35}$ is34.9800U; Mass of Proton ($_1p_1$) is 1.008665U and Mass of neutron ($_0n^1$)is1.007825U
(c) Show that radioactive decay follows exponential law.
(d) What are the devices used for measuring the solar radiations? Explain with any one of them with neat sketch.
(e) With neat sketch explain the concept of green building .Also write the factors which can made a building green?

SECTION C

3. Attempt any *one* part of the following: 10x1=10

(a) Explain the Carnot vapour power cycle with T-s diagram. Also find out the efficiency of Carnot cycle.
(b) Explain Brayton cycle and obtained expression for efficiency in terms of pressure and temperature ration.

4. Attempt any *one* part of the following: 10x1=10

(a) What do you mean by binding energy? What is the total binding energy per nucleon for the $_6C^{12}$ nucleus?
(b) With a neat sketch, explain pressurized water reactor (PWR) highlights its merits and demerits.

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5. Attempt any *one* part of the following: 10x1=10

(a) Discuss p-n junction in forward bias and reversed bias condition.
(b) Derive V-I and P-V characteristics of photo volatile device.

6. Attempt any *one* part of the following: 10x1=10

(a) Derive the continuity equation for 1-D fluid flow. Also derive the expression for wind power.
(b) Write down the principle of power generation in wind mills. Derive an expression for maximum efficiency.

7. Attempt any *one* part of the following: 10x1=10

(a) Describe steps for methodology for energy audit. List the key instrument for energy audit.
(b) What is a nuclear fuel cycles? What steps are involved in Nuclear fuel cycle?

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